Def V.1.9. Take $\mathfrak{l}$ as an ideal in A. $G = A$. $G_n = \mathfrak{l}^n$. then the topology defined on A is called the $\mathfrak{l}$-adic topology.

Remark. 1). $\mathfrak{l}$-adic topology is Hausdorff $\Leftrightarrow \bigcap \mathfrak{l}^n = (0)$.

2) $\phi: A \to \hat{A}$ is a cantinuous ring homomorphism, $\ker \phi = \bigcap \mathfrak{l}^n$.

Def V.1.10. M is an A-module. $G = M$. $G_n = \mathfrak{l}^n M$. This defines the $\mathfrak{l}$-topology on M.

Lemma V.1.11. (Hensel Lemma) Let (R, m) be a ~~m-adic~~ local ring that is complete with respect to m-adic topology, $f(x) \in R[x]$, Suppose $\exists\, x_0 \in R$ s.t. $\begin{cases} f(x_0) \equiv 0 \pmod m \\ f'(a_0) \notin m \ (f'(a_0) \text{ is a unit in } R) \end{cases}$.

Then $\exists!\ a \in R$. s.t. $f(a) = 0$. $a \equiv a_0 \pmod m$

pf: 1. Existence: we construct a sequence $\{a_n\}_n \subseteq R$. s.t.

(i) $a_n \equiv a_0 \pmod m$.

(ii) $f(a_n) \in m^{n+1}$

(iii) $a_{n+1} \equiv a_n \pmod{m^{n+1}}$.

$n=0$ ✓. Assuming that we have constructed a_n

$\Rightarrow a_n \equiv a_0 \pmod m \Rightarrow f'(a_n) \equiv f'(a_0) \not\equiv 0 \pmod m$

Let $v = f'(a_n)^{-1}$ and let $a_{n+1} = a_n - f(a_n)v$. $\Rightarrow a_{n+1} \equiv a_0 \pmod m$ (i).

$\Rightarrow a_{n+1} - a_n = -f(a_n)v \in m^{n+1}$ (iii).

By ~~Taylor~~ Taylor expansion over a ring:

we have $f(a_{n+1}) = f(a_n) + f'(a_n)(-f(a_n)v) + h^2 g(a_n, h) = h^2 g(a_n, h)$.

where $h = a_{n+1} - a_n \in m^{n+1} \Rightarrow h^2 \in m^{2n+2} \in m^{n+2}$. $\Rightarrow f(a_{n+1}) \in m^{n+2}$ (ii)

2. Convergence: $\forall\, m > n \geq 0$. we have $a_m - a_n = \sum_{i=n}^{m-1} (a_{i+1} - a_i) \in m^{n+1}$

$\Rightarrow \{a_n\}$ is a Cauchy sequence.

By the completeness. $\exists!\ a$ s.t. $a \equiv a_n \pmod{m^{n+1}}$. $\forall\, n$. $(a \equiv a_0 \bmod m)$

The poly-map is continuous $f(a) = f(\lim_{n\to\infty} a_n) = \lim_{n\to\infty} f(a_n) = 0$.

$\Rightarrow f(a) = 0$.

3. Uniqueness: $f(a) = f(b) = 0$ $\quad a \equiv b \equiv a_0 \pmod m$. $d = a - b \in m$.

$\Rightarrow 0 = f(a) = f(b) + f'(b)d + d^2 g(b, d)$ $(d = a - b \in m)$.

$\Rightarrow d = -d^2 g(b, d) f'(b)^{-1} \Rightarrow d(1 - dh) = 0$. $(d \in m)$

$\Rightarrow dh \in m$. $\Rightarrow 1 - dh$ is a unit in $R \Rightarrow d = 0 \Rightarrow a = b$ ▨

VI. Primary Decomposition (the alternative of prime's powers in a common ring)

1. Basic definitions and properties.

Def. VI.1.1. $\mathfrak{q} \subsetneq A$ is an ideal, then $\mathfrak{q}$ is primary iff $xy \in \mathfrak{q} \Rightarrow x \in \mathfrak{q}$ or $y^n \in \mathfrak{q}$.

Remark. $\mathfrak{q}$ is primary $\iff$ $A/\mathfrak{q} \neq 0$, any zero-divisor in $A/\mathfrak{q}$ is nilpotent

Prop. VI.1.2. $\mathfrak{q}$ is a primary ideal in A, then $\sqrt{\mathfrak{q}}$ is the smallest prime ideal containing $\mathfrak{q}$.

pf: By the previous propsition, $\sqrt{\mathfrak{q}} = \bigcap_{\mathfrak{q} \subseteq \mathfrak{p}_i} \mathfrak{p}_i := \mathfrak{p} \supseteq \mathfrak{q}$.

$\forall\, xy \in \mathfrak{p} \Rightarrow (xy)^m \in \mathfrak{q} \Rightarrow x \in \sqrt{\mathfrak{q}}$ or $\sqrt{\mathfrak{q}}$ thus $\mathfrak{p}$ is a prime ideal ▨.

Remark. $\mathfrak{q}$ is said to be $\mathfrak{p}$-primary, if $\mathfrak{q}$ is primary and $\sqrt{\mathfrak{q}} = \mathfrak{p}$.

Examples. 1). The primary ideals in $\mathbb{Z}$ are (0) and (p^n) where p is prime,

2). $A = k[x,y]$, $\mathfrak{q} = \langle x, y^2 \rangle$ Then $\sqrt{\mathfrak{q}} = \langle x, y \rangle$, $(A/\mathfrak{q} \cong k[y]/(y^2) \Rightarrow \mathfrak{q}$ is primary

$\mathfrak{p}^2 \subsetneq \mathfrak{q} \subsetneq \mathfrak{p}$.

Prop. VI.1.3. If $\sqrt{\mathfrak{a}}$ is maximal, then $\mathfrak{a}$ is primary. The powers of a maximal ideal $\mathfrak{m}$ is $\mathfrak{m}$ primary.

pf: let $\sqrt{\mathfrak{a}} = \mathfrak{m}$., then $\sqrt{A/\mathfrak{a}} = \bigcap_{\mathfrak{p} \in A/\mathfrak{a}} \mathfrak{p}$ $\mathrm{nil}(A/\mathfrak{a}) = \pi(\sqrt{\mathfrak{a}})$. is maximal.

$\Rightarrow$ $A/\mathfrak{a}$ has only one prime ideal. $\Rightarrow$ every zero-divisor in nilpotent. in $A/\mathfrak{a}$ ▨

Lemma VI.1.4. $\mathfrak{q}_i$ $(1 \le i \le n)$ are $\mathfrak{p}$-primary, then $\mathfrak{q} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ are $\mathfrak{p}$-primary.

pf: $\sqrt{\mathfrak{q}} = \sqrt{\bigcap_{i=1}^{n} \mathfrak{q}_i} = \bigcap_{i=1}^{n} \sqrt{\mathfrak{q}_i} = \mathfrak{p}$.; $\forall\, xy \in \mathfrak{q}$, $y \notin \mathfrak{q} \Rightarrow \exists\, i$. s.t. $xy \in \mathfrak{q}_i$, $y \notin \mathfrak{q}_i$

$\Rightarrow x \in \mathfrak{p}$. ▨

Lemma VI.1.5. $\mathfrak{q}$ be a $\mathfrak{p}$-primary ideal. $x \in A$. Then

i). if $x \in \mathfrak{q}$ then $(\mathfrak{q} : x) = (1)$

ii) if $x \notin \mathfrak{q}$ then $(\mathfrak{q} : x)$ is $\mathfrak{p}$-primary, therefore $\sqrt{(\mathfrak{q} : x)} = \mathfrak{p}$

iii). if $x \notin \mathfrak{p}$ then $(\mathfrak{q} : x) = \mathfrak{q}$.

pf: i) & iii) is obvious.

ii) $y \in (\mathfrak{q} : x) \iff xy \in \mathfrak{q} \Rightarrow y \in \mathfrak{p}$. $\Rightarrow \mathfrak{q} \subseteq (\mathfrak{q} : x) \subseteq \mathfrak{p}$. $\Rightarrow \sqrt{(\mathfrak{q} : x)} = \mathfrak{p}$.

$yz \in (\mathfrak{q} : x) \Rightarrow xyz \in \mathfrak{q}$. $\Rightarrow xz \in \mathfrak{q} \Rightarrow z \in (\mathfrak{q} : x)$ ▨

$y \notin \mathfrak{p}$ (because $\mathfrak{q}$ is $\mathfrak{p}$-primary)

Def VI.1.6. $\mathfrak{a} \subseteq A$ Suppose that $\mathfrak{a}$ has a primary decomposition:

$$\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i \quad \sqrt{\mathfrak{q}_i} \neq \sqrt{\mathfrak{q}_j}, \forall i \neq j \quad \text{and} \quad \mathfrak{q}_j \not\supseteq \bigcap_{i \neq j} \mathfrak{q}_i$$

Then we call $\mathfrak{a}$ is decomposable.

Thm VI.1.6. (Uniqueness thm) $\mathfrak{a} \subseteq A$ is ~~dep~~ decomposable, $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ minimal. $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$.

Then $\{\mathfrak{p}_i \mid 1 \le i \le n\} = \{\mathfrak{p} \mid \mathfrak{p} = \sqrt{(\mathfrak{a}:x)}, x \in A\}$. (*)

pf: $(\mathfrak{a}:x) = \bigcap_{i=1}^{n} (\mathfrak{q}_i : x) \Rightarrow \sqrt{(\mathfrak{a}:x)} = \bigcap_{i=1}^{n} \sqrt{(\mathfrak{q}_i:x)} \overset{(4.5)}{=\!=} \bigcap_{x \notin \mathfrak{q}_j} \mathfrak{p}_j$

① $\sqrt{(\mathfrak{a}:x)}$ is prime $\Rightarrow \exists j$ s.t. $\sqrt{\mathfrak{a}:x} = \mathfrak{p}_j$ (Prime avoidance):

② $\forall i \in [1,n]$, $\exists x_i$. s.t. $x_i \notin \mathfrak{q}_i$ ➡, $x_i \in \bigcap_{j \neq i} \mathfrak{q}_j \Rightarrow \sqrt{(\mathfrak{a}:x_i)} = \mathfrak{p}_i$ ▨

Remark. 1) $(\mathfrak{a}:x_i) = (\mathfrak{q}_i : x)$ is $\mathfrak{p}_i$-primary.

2). $A/\mathfrak{a}$ is an A-module, then $(\mathfrak{a}:x_i) = \mathrm{Ann}_{A/\mathfrak{a}}(x_i)$

3). We call $\mathfrak{p}_i$ in (*) is associated with $\mathfrak{a}$, the minimal elements in $\{\mathfrak{p}_i \mid 1 \le i \le n\}$. is called the minimal/isolated prime ideals belonging to $\mathfrak{a}$, the others are called embedded prime ideals

Examples: 1) $\mathfrak{a} = (x^2, xy)$. $A = k[x,y]$ Then $\mathfrak{a} = \mathfrak{p}_1 \cap \mathfrak{p}_2^2$ where $\mathfrak{p}_1 = (x)$, $\mathfrak{p}_2 = (x,y)$.

Then $\mathfrak{p}_1 \subsetneq \mathfrak{p}_2$ ∴, $\mathfrak{p}_1$ is the isolated ideal. $\mathfrak{p}_2$ is the embedded ideal.

2).

Prop. VI.1.7. $\mathfrak{a}$ decomposable, then $\{\mathfrak{p} \mid \mathfrak{p} \supseteq \mathfrak{a}, \mathfrak{p}$ minimal$\}$ $\overset{=}{\text{➡}}$ $\{\mathfrak{p} \mid \mathfrak{p}$ is the minimal/isolated ideal belonging to $\mathfrak{a}\}$
$\quad\quad\quad

Remark. $\mathcal{D} = \bigcup$ all prime ideals belonging to 0;

$R = \mathrm{nil}(A) = \bigcap$ all minimal prime ideals belonging to 0

(If the zero ideal is decomposable)

Prop. VI.1.9 (Primary ideals under localization). S is a multiplicatively closed subset of A, $\mathfrak{q}$ be a $\mathfrak{p}$-primary ideal.

Then i). $S \cap \mathfrak{p} \neq \phi \Rightarrow S^{-1}\mathfrak{q} = S^{-1}A$

ii) $S \cap \mathfrak{p} = \phi \Rightarrow S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$-primary, and $(S^{-1}\mathfrak{q})^c = \mathfrak{q}$.

pf: i) is obvious.

ii). $S \cap \mathfrak{p} = \phi \Rightarrow \forall s \in S.\ x \in A,\ sx \in \mathfrak{q} \Rightarrow x \in \mathfrak{q}$ $\Rightarrow$ let $x/1 \in S^{-1}\mathfrak{q} \Rightarrow \exists t \in S$ s.t. $tx \in \mathfrak{q} \Rightarrow x \in \mathfrak{q}$

$\Rightarrow \mathfrak{q}^{ec} = \mathfrak{q}$, Also $\sqrt{S^{-1}\mathfrak{q}} = S^{-1}\mathfrak{p}$. $\Rightarrow S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$-primary ▨

Notation: let $S(\mathfrak{a})$ be the contraction of $S^{-1}\mathfrak{a}$

Prop. VI.1.10. let $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ minimal, $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$. And $\mathfrak{p}_i \cap S = \phi$ $(1 \le i \le m)$ $\mathfrak{p}_i \cap S \neq \phi$ (

Then $S^{-1}\mathfrak{a} = \bigcap_{i=1}^{m} S^{-1}\mathfrak{q}_i$. $S(\mathfrak{a}) = \bigcap_{i=1}^{m} \mathfrak{q}_i$.

pf: (skipped).

Remark. A set Σ of prime ideals belonging to $\mathfrak{a}$ is isolated iff.: if $\mathfrak{p}'$ belongs to $\mathfrak{a}$ and $\mathfrak{p}' \supseteq \mathfrak{p}$ for some $\mathfrak{p} \in \Sigma$. then $\mathfrak{p}' \in \Sigma$.

Let $S = A - \bigcup_{\mathfrak{p} \in \Sigma} \mathfrak{p}$, then S is a multiplicatively ~~isolated~~ closed set.

$\mathfrak{p}' \in \Sigma \Rightarrow \mathfrak{p}' \cap S = \phi$.

$\mathfrak{p}' \notin \Sigma \Rightarrow \mathfrak{p}' \nsubseteq \bigcup_{\mathfrak{p} \in \Sigma} \mathfrak{p}$ (Prime avoidance) $\Rightarrow \mathfrak{p}' \cap S \neq \phi$

By (VI.1.10) We have:

Thm VI.1.11. (Uniqueness thm) $\mathfrak{a}$ decomposable, $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ minimal, Σ is an isolated set of prime ideals of $\mathfrak{a}$. Then $\bigcap_{\mathfrak{p} \in \Sigma}$ [illegible] is independent of decomposition.

pf: (skipped).

Cora VI.1.12. The isolated primary components are uniquely determined by $\mathfrak{a}$.

pf: $\forall\ \mathfrak{q}_i$ corresponds to minimal primes ideals $\mathfrak{p}_i$. Take $\Sigma = \{\mathfrak{p}_i\}$ ▨

VII. Integral Dependence.

1. Integral Dependence.

Def VI.1.1. $A \subseteq B$ subring. $x \in B$ is integral over A if x is a root of a monic polynomial with coefficients in A.

Def VI.1.2. $A \subseteq B$ subring. $A \subseteq C \subseteq B$ $C := \{b \in B \mid b \text{ is integral over } A\}$

Then C is called the closed closure of A in B.

1) If $C = A$, then call A integrally closed.

2) If $C = B$, then call that B is integral over A.

3) A integrally closed $\Leftrightarrow$ A is an integral domain & A is integrally closed in $\mathrm{Frac}(A)$

Prop. VII 1.3. $A \subseteq B$, TFAE:

(1) $x \in B$ is integral over A;

(2) $A[x]$ is a finitely generated A-module.

(3) $\exists$ subring $C \subseteq B$. s.t. $A[x] \subseteq C \subseteq B$. and C is a finitely generated A-module.

(4). There exists a faithful $A[x]$-module M $(\mathrm{Ann}_{A[x]} M = 0)$. which is finitely generated as an A-module.

pf: (1). $\Rightarrow$ (2). $x \in B$ integral over A. $\exists a_1, \dots, a_n \in A$. s.t. $x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n = 0$.

$\Rightarrow A[x]$ is generated by $1, x, \dots, x^{n-1}$

(2) $\Rightarrow$ (3). Take $C = A[x]$.

(3) $\Rightarrow$ (4). Take $M = C$. then $yC = 0 \Rightarrow y \cdot 1 = 0$ thus M is a faithful $A[x]$-module.

(4) $\Rightarrow$ (1). Consider the $A[x]$-module homomorphism $\phi: M \to M$ $m \mapsto x \cdot m$.

M is fg-A-mod, By Caley-Hamilton thm, we have $\exists a_1, \dots, a_n \in A$, s.t. $x^n + a_1 x^{n-1} + \dots + a_n = 0$ in $\mathrm{End}_A(M)$

And from M is a faithful, we have $x^n + a_1 x^{n-1} + \dots + a_n = 0$ in $A[x]$ ▨

Cora. VII. 1.4. x_i $(1 \le i \le n) \in B$ is integral over A. Then $A[x_1, \dots, x_n]$ is a finitely-generated A-algebra

pf: By induction on n (skipped) ▨

Cora VII. 1.5. The integral closure of A in B is a subring of B containing A.

pf: If $x, y \in \overline{A}$. then $A[x, y]$ is a finitely-generated A-module.

$\Rightarrow A[x \pm y] \subseteq A[x, y]$, $A[xy] \subseteq A[x, y]$ $\overset{(3)}{\Rightarrow}$ $x \pm y, xy \in \overline{A}$. ▨

Example: The integral closure of $\mathbb{Z}$ in the algebraic number field L. is called the ring of integers $\mathcal{O}_L$ in L.

Def VII.1.6. $f: A \to B$ ring homomorphism (B is an A-algebra).

Then f is integral and B is an integral A-algebra.

iff B is integral over $f(A)$

Cora VII.1.7. f is integral, and finite-type, then f is finite.

Cora VII.1.8. $A \subseteq B \subseteq C$ subrings, If B integral over A, C integral over B. Then C integral over A

pf: By restrictions of scalars ▨

Cora VII.1.9. $A \subseteq B$. C is the integral closure of A in B. then C is integrally closed in B.

pf: $x \in B$ integral over $C \Rightarrow x$ integral over $A \Rightarrow x \in C$ ▨

2. Going-up property.

Prop. VII.2.1. $A \subseteq B$ rings. B integral over A.

i). $\mathfrak{b} \subseteq B$ ideal. $\mathfrak{a} = \mathfrak{b}^c = A \cap \mathfrak{b}$. then $B/\mathfrak{b}$ integral over $A/\mathfrak{a}$.

ii). $S^{-1}B$ integral over $S^{-1}A$.

pf: i) $x^n + a_1 x^{n-1} + \dots + a_n = 0 \pmod{\mathfrak{b}}$. ▨

ii) $\frac{x}{s} \in S^{-1}B \Rightarrow x^n + a_1 x^{n-1} + \dots + a_n = 0 \Rightarrow (\frac{x}{s})^n + \dots + a_n = 0$ ▨

Prop. VII.2.2. $A \subseteq B$ integral domains. B integral over A. Then B is a field iff A is a field

pf: $\Leftarrow$: $0 \neq b \in B$. $\Rightarrow b^n + a_1 b^{n-1} + \dots + a_n = 0$, $(a_n \neq 0)$.

$\Rightarrow b^{-1} = -a_n^{-1}(b^{n-1} + a_1 b^{n-2} + \dots + a_{n-1})$

$\Rightarrow$: $0 \neq x \in A \subseteq B \Rightarrow x^{-1} \in B \Rightarrow (x^{-1})^m + b_1 (x^{-1})^{m-1} + \dots + b_m = 0$

$\Rightarrow x^{-1} = -(b_1 + b_2 x + \dots + b_m x^{m-1}) \in A$ ▨

Prop. VII.2.3. $f: A \to B$ integral. $\mathfrak{q} \in \operatorname{Spec} B$. $\mathfrak{p} = \mathfrak{q} \cap A \in \operatorname{Spec} A$. Then $\mathfrak{q}$ maximal $\Leftrightarrow$ $\mathfrak{p}$ maximal.

~~In other words $f^*: \operatorname{Spec} B \to \operatorname{Spec} A$ is the closed mapping.~~

pf: By VII.2.1.i). $B/\mathfrak{q} \leftarrow A/(\mathfrak{q} \cap A)$ integral. and $A/(\mathfrak{q} \cap A)$, $B/\mathfrak{q}$ are integral domains

$\Rightarrow$: $\mathfrak{q}$ maximal $\Leftrightarrow$ $\mathfrak{p}$ maximal.

Prop 2.4: $\mathfrak{q}, \mathfrak{q}' \subseteq B$ prime, s.t. $\mathfrak{q} \subseteq \mathfrak{q}'$ and $\mathfrak{q}^c = \mathfrak{q}'^c = \mathfrak{p}$ Then $\mathfrak{q} = \mathfrak{q}'$.

pf: $B_{\mathfrak{p}}$ integral over $A_{\mathfrak{p}}$

$$\begin{array}{ccc} A & \longrightarrow & A_{\mathfrak{p}} \\ \downarrow & & \downarrow \\ B & \longrightarrow & B_{\mathfrak{p}} \end{array}$$

we have $\mathfrak{p}^e$ in $A_{\mathfrak{p}}$ is maximal.

and $\mathfrak{q}^e, \mathfrak{q}'^e$ in $B_{\mathfrak{p}} \Rightarrow \mathfrak{q}^e \subseteq \mathfrak{q}'^e$.

$\Rightarrow$ $\mathfrak{q}^e \cap A_{\mathfrak{p}} = \mathfrak{q}'^e \cap A_{\mathfrak{p}} = \mathfrak{p}^e$ maximal $\Rightarrow$ $\mathfrak{q}^e$, $\mathfrak{q}'^e$ maximal. $\Rightarrow \mathfrak{q} = \mathfrak{q}'$ ▨

Thm VII.2.5. $f: A \to B$ integral. Then $f^*: \operatorname{Spec} B \to \operatorname{Spec} A$ is surjective.

pf: $\forall \mathfrak{p} \in \operatorname{Spec} A$. $\Rightarrow B_{\mathfrak{p}}$ integral over $A_{\mathfrak{p}}$ Then $\mathfrak{p}A_{\mathfrak{p}}$ maximal.

$$\begin{array}{ccc} B & \xrightarrow{\beta} & B_{\mathfrak{p}} \supseteq m \\ \uparrow & & \uparrow \\ A & \xrightarrow{\alpha} & A_{\mathfrak{p}} \\ \mathfrak{p} & & m \cap A_{\mathfrak{p}} \end{array}$$

choose $m \subseteq B_{\mathfrak{p}}$ maximal. By (2.3) we have $m \cap A_{\mathfrak{p}}$ maximal in $A_{\mathfrak{p}}$.

Then By $A_{\mathfrak{p}}$ is a local ring, we have $m \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$

Then let $\mathfrak{q} = \beta^{-1}(m) \in \operatorname{Spec} B \Rightarrow \mathfrak{q} \cap A = \alpha^{-1}(\mathfrak{p}A_{\mathfrak{p}}) = \mathfrak{p}$ ▨

Thm VII.2.6. (Going-up theorem) $A \subseteq B$ rings. B integral over A.

Let $\mathfrak{q}_1 \subsetneq \dots \subsetneq \mathfrak{q}_m \in \operatorname{Spec} B$ and

$\mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_m \subsetneq \dots \subsetneq \mathfrak{p}_n$ $(n>m) \in \operatorname{Spec} A$. s.t. $\forall$ $1 \le i \le m$, $\mathfrak{q}_i \cap A = \mathfrak{p}_i$

Then $\exists\ \mathfrak{q}_{m+1}, \dots, \mathfrak{q}_n \in \operatorname{Spec} B$. s.t. $\mathfrak{q}_1 \subsetneq \mathfrak{q}_2 \subsetneq \dots \subsetneq \mathfrak{q}_n$ & $\mathfrak{q}_i \cap A = \mathfrak{p}_i$, $\forall i$

pf: By induction: $m=1$. $n=2$. we have $\overline{A} = A/\mathfrak{p}_1$. $\overline{B} = B/\mathfrak{q}_1$.

$\overline{A} \subseteq \overline{B}$. and $\overline{B}$ integral over $\overline{A}$.

Then by (2.5) we have $\exists\ \overline{\mathfrak{q}_2} \in \operatorname{Spec} \overline{B}$. s.t $\overline{\mathfrak{q}_2} \cap A = \overline{\mathfrak{p}_2}$. Then lift back $\overline{\mathfrak{q}_2}$.

We have $\mathfrak{q}_2 = \pi^{-1}(\overline{\mathfrak{q}_2})$ $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$ ▨

3. Going down Property.

Prop. VII.3.1. (Local property): Let A be an integral domain. TFAE:

i) A is integrally closed

ii) $A_{\mathfrak{p}}$ is ... $\forall \mathfrak{p}$ prime

iii) $A_{\mathfrak{m}}$ is ... $\forall \mathfrak{m}$ maximal.

Lemma: $A \subseteq C \subseteq B$. C is the integral closure of A in B. Then $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$

pf: $S^{-1}C$ integral over $S^{-1}A$, let $\frac{x}{s} \in S^{-1}B$ integral over $S^{-1}A$

Then $(\frac{x}{s})^n + \frac{a_1}{s_1}(\frac{x}{s})^{n-1} + \dots + \frac{a_n}{s_n} = 0 \Rightarrow (s_1 s_2 \dots s_n x) \in C \Rightarrow \frac{x}{s} \in S^{-1}C$ ▨

Back to the proof: $K = \operatorname{Frac}(A)$, C be the integral closure of A in K.

$f: A \to C$ $x \mapsto x$. Then A integrally closed $\Leftrightarrow f$ surjective.

$A_{\mathfrak{p}}$ ($A_{\mathfrak{m}}$) integrally closed $\Leftrightarrow f_{\mathfrak{p}}$ ($f_{\mathfrak{m}}$) surjective. ▨

Def VII.3.2 $A \subseteq B$. $\mathfrak{a} \subseteq A$ ideal, $x \in B$ integral over $\mathfrak{a}$ iff $\exists a_i \in I$. s.t. $x^n + a_1 x^{n-1} + \dots + a_n = 0$.

Lemma VII.3.3. $\mathfrak{a} \subseteq A \subseteq C \subseteq B$. C be the closure, $\mathfrak{a}$. Then the closure of $\mathfrak{a}$ in $B = \sqrt{\mathfrak{a}^e}$. $\mathfrak{a}^e : A \to C$.

pf: ① $x \in B$ integral over $\mathfrak{a}$. $\Rightarrow$ $x^n + a_1 x^{n-1} + \dots + a_n = 0$ $a_i \in \mathfrak{a} \subseteq C$.

$\Rightarrow$ $x \in C$, $x^n \in \mathfrak{a}^e \Rightarrow x \in \sqrt{\mathfrak{a}^e}$.

② $x \in \sqrt{\mathfrak{a}^e}$, $\Rightarrow x^n \in \mathfrak{a}^e$. that is $x^n = \sum_{i=1}^{m} a_i x_i$ $a_i \in \mathfrak{a}$, $x_i \in C$

where x_i integral over A. $\Longrightarrow$ $A[x_1, \dots, x_m]$ is a finite generated A-algebra M.

and $x^n M \subseteq \mathfrak{a} M$

Consider $\phi: M \to M$ $m \mapsto x^n \cdot m$. By Caley-Hamilton thm, $\phi(x^n)^r + b_1(x^n)^{r-1} + \dots + b_r = 0$

in $\operatorname{End}_A(M)$. $\phi(M) \subseteq \mathfrak{a} M \Rightarrow$ $(x^n)^r + b_1(x^n)^{r-1} + \dots + b_r = 0$. in A $(b_i \in A)$

$\Rightarrow x^n$ integral over $\mathfrak{a}$ $\Rightarrow$ x integral over $\mathfrak{a}$ ▨

Prop. VII.3.4. $A \subseteq B$ integral domains, A integrally closed, and let $x \in B$ be integral over $\mathfrak{a} \subseteq A$.

Then x is algebraic over Frac$(A) = K$. ①.

And if $\mathrm{Irr}_K(x) = t^n + a_1 t^{n-1} + \cdots + a_n \in K[t]$, then $a_i \in \sqrt{\mathfrak{a}}$. ②

(Particularly, $\mathfrak{a} = (1) \Rightarrow a_i \in A$) (**)

pf: ① obvious; ②: L be the splitting field of $\mathrm{Irr}_K(x)$ over K. $L = K(x_1, \ldots, x_n)$.

x_i integral over $\mathfrak{a}$. $(\forall \sigma \in \mathrm{Gal}(\bar{K}/K), \sigma|_K = \mathrm{id}\}, \sigma|_{(x_1,\ldots,x_n)} \subseteq S_n)$

By Vieta thm. a_i are polynomials in x_i. $\overset{(3.3)}{\Rightarrow}$ a_i integral over $\mathfrak{a}$.

~~where~~. Since A integrally closed $\overset{(3.3)}{\Rightarrow}$ $a_i \in \sqrt{\mathfrak{a}}$. ▨

Thm (Going-down thm). $A \subseteq B$, B integral over A., A integral closed.

let $\mathfrak{q}_1 \supsetneq \cdots \supsetneq \mathfrak{q}_m \in \mathrm{Spec}\, B$.

$\mathfrak{p}_1 \supsetneq \cdots \supsetneq \mathfrak{p}_m \supsetneq \cdots \supsetneq \mathfrak{p}_n \in \mathrm{Spec}\, A$ $(n > m)$.

s.t. $\mathfrak{q}_i \cap A = \mathfrak{p}_i$. $(1 \le i \le m)$

Then $\exists\, \mathfrak{q}_{m+1} \supsetneq \cdots \supsetneq \mathfrak{q}_n$ s.t. $\mathfrak{q}_1 \supsetneq \cdots \supsetneq \mathfrak{q}_n$ ~~&~~ $\mathfrak{q}_i \cap A = \mathfrak{p}_i$ $(\forall 1 \le i \le n)$

pf: $m = 1$, $n = 2$. find $\mathfrak{q}_2 \subsetneq \mathfrak{q}_1$. s.t. $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$.

$\mathfrak{p}_2$ $A \xrightarrow{f} B \xrightarrow{\mathfrak{p}_2 B}$ We prove that: $B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A = \mathfrak{p}_2$.

$\beta \downarrow$

$B_{\mathfrak{q}_1}$ $\mathfrak{p}_2 B_{\mathfrak{q}_1}$ $\forall x \in B_{\mathfrak{q}_1}\mathfrak{p}_2$. where $x = \frac{y}{s}$. $y \in \mathfrak{p}_2 B$, $s \in B - \mathfrak{q}_1$

Then y is integral over $\mathfrak{p}_2$. (3.3). then its minimal equation over Frac(A) is

$y^r + u_1 y^{r-1} + \cdots + u_r = 0$. $(u_i \in \mathfrak{p}_2)$.

$\forall x \in B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A$, Then $s = yx^{-1}$ with x^{-1} in K $\Rightarrow s^r + v_1 s^{r-1} + \cdots + v_r = 0$ $v_i = \frac{u_i}{x^i}$

$\Rightarrow x^i \cdot v_i \in \mathfrak{p}_2$. ●

Since s integral over A, By (**) we have $v_i \in A$.

If $x \notin \mathfrak{p}_2 \Rightarrow v_i \in \mathfrak{p}_2 \Rightarrow s^r \in B\mathfrak{p}_2 \subseteq B\mathfrak{q}_1 \subseteq \mathfrak{q}_1 \Rightarrow s \in \mathfrak{q}_1$ contradiction

$\Rightarrow x \in \mathfrak{p}_2$. $\Rightarrow$ $B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A = \mathfrak{p}_2$.

Now take $\mathfrak{q}_2 = \beta^{-1}(B_{\mathfrak{q}_1}\mathfrak{p}_2)$ we have $\mathfrak{q}_2 \subsetneq \mathfrak{q}_1$, $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$ ▨